HOPE MOCAP BRINGUP

P1-to-Pelvis Calibration Walkthrough

A high-level guide to aligning the OptiTrack P1 marker-cluster pose with the Agibot A3 pelvis root frame.

CALIBRATION OUTCOME

world -> P1 -> pelvis_link

Live mocap pose plus one saved static transform

Scope: one-time marker-shell bringup, runtime TF publication, and simulator/controller boundaries. Technical commands and parameter details remain in the repository READMEs.

Why this calibration exists

The motion-capture system and the robot describe the same moving body from two different coordinate frames. Calibration records their fixed geometric relationship once, then reuses it at runtime.

world	→	P1	→	pelvis_link
arena reference		marker-cluster frame		A3 root link

P1 is what mocap sees
Motive solves a 6-DOF rigid body from the pelvis marker shell. Its default pivot and axes generally do not coincide with the robot root.

pelvis_link is what robot software names
A3 simulation and robot models use pelvis_link as the physical root body. That frame should remain the endpoint of the calibrated TF.

Important naming rule: /a3/calibration/pelvis_pose is a temporary PoseStamped topic. It is not a TF frame named pelvis_pose.

Why the A3 pelvis IMU is not enough

The checked-in hardware bridge provides pelvis orientation and inertial measurements, but not absolute translation in the arena. A full 6-DOF reference is therefore required during the calibration session.

Verified P1 marker shell: all ten points

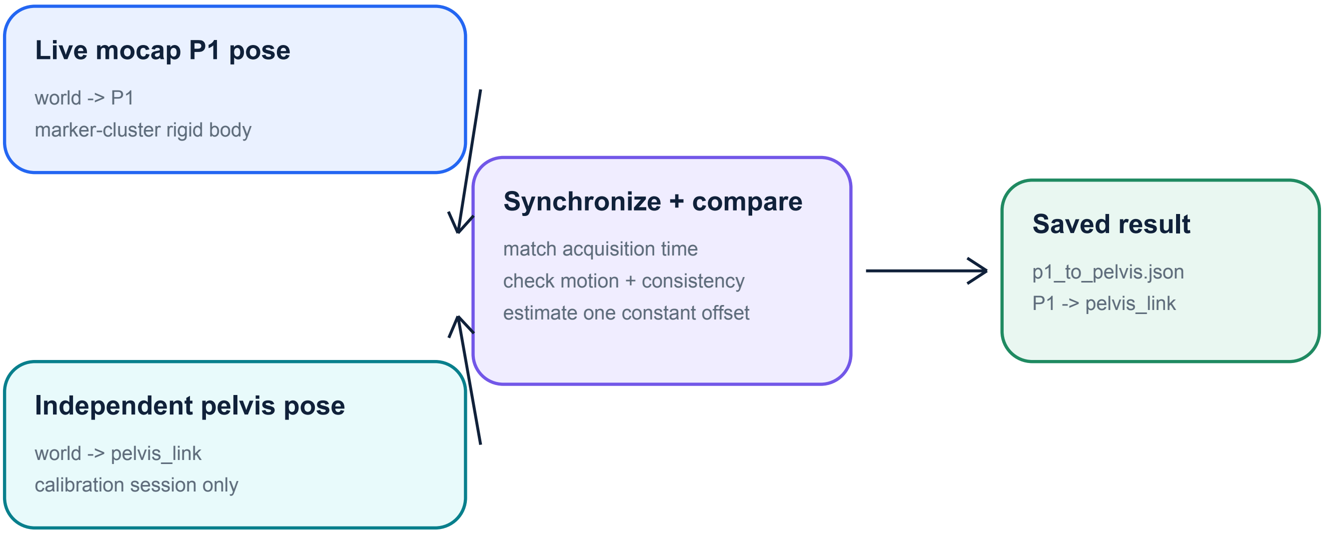
Motive P1 rigid body: f1-f5 and b1-b5. The 0702 shell layout and physical mocap experiment confirm that the complete ten-point set is available and visible.

CAD centroid in pelvis_link	Axis-aligned pivot cross-check
[-0.0024, 0, -0.1490] m	[+2.4, 0, +149.0] mm

Use all ten markers by default. These values are a translation cross-check only; the synchronized live calibration remains authoritative for the installed 6-DOF P1 -> pelvis_link transform. See mocap/README.md and docs/OPTITRACK.md.

What happens during calibration

Two independent systems observe the pelvis at matching times. The calibrator compares their poses and extracts the transform that stays constant as the robot moves.



Input A: mocap
The live P1 rigid-body pose from Motive: world -> P1.

Input B: reference
An independent full pelvis pose: world -> pelvis_link. This source exists only for calibration.

Output: record
A quality-checked JSON containing the constant P1 -> pelvis_link transform.

The reference must not be calculated from P1. Doing so would make the calibration circular and unable to reveal the actual offset.

Frame and timing rule: both input poses must use the same reference frame and a shared acquisition-time epoch. A topic name alone does not establish either condition.

One-time operator walkthrough

Perform this sequence for each installed marker shell and repeat it whenever the Motive P1 rigid-body definition or mounting changes.

1. **Prepare the two pose sources**
   Start the OptiTrack P1 stream and an independent external tracker or state estimator that reports the A3 pelvis in the same arena frame.

2. **Confirm frames, clocks, and independence**
   Verify that both streams represent the same physical time and that the pelvis reference is not derived from P1 or from the saved static transform.

3. **Collect an excited motion sequence**
   Move the pelvis smoothly through translation and rotation. A stationary capture can look repeatable while providing too little information.

4. **Review the quality gates**
   Check synchronized sample count, timestamp behavior, motion span, pair skew, rejected outliers, and residual consistency.

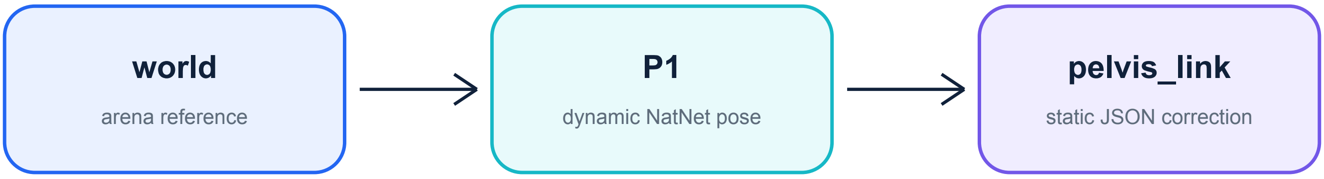
5. **Save and identify the result**
   Store p1_to_pelvis.json with the robot, marker shell, Motive asset/profile, date, and operator session so the setup can be reproduced.

6. **Switch to normal runtime**
   Stop the independent calibration reference. Load the JSON and publish the constant P1 -> pelvis_link transform alongside the live P1 mocap pose.

The Table rigid body is not an input to this calibration and remains absent from the competition stream.

Normal runtime after calibration

The temporary full-pelvis reference is no longer required. The live P1 pose and the saved constant offset together provide the pelvis pose in the arena.



Changes continuously
world -> P1 follows the robot in real time and comes from the mocap rigid-body stream.

Stays constant
P1 -> pelvis_link comes from the saved calibration and remains fixed while the marker shell stays rigidly mounted.

Optional Motive pivot correction

The measured translation and rotation may instead be absorbed into Motive's P1 pivot definition. Recalibration should then report an approximately identity correction.

Choose exactly one correction path: Motive-corrected P1 output OR the ROS static P1 -> pelvis_link publisher. Never apply both.

What this enables

ROS planners, logging, visualization, and downstream integrations can resolve an absolute world -> pelvis_link pose. The native A3 controller does not automatically consume this TF; an explicit adapter is required if a controller needs the absolute pelvis pose.

Simulator and controller boundaries

The same physical root is represented differently across the stack. Keep the link/frame semantics separate from ROS topic names.

Environment	What it uses	Calibration implication
Isaac Lab	Root rigid body pelvis_link and simulator state tensors.	No ROS pelvis_pose topic is required for training.
MuJoCo	Body pelvis_link; simulation publishes a pose topic in odom.	Useful for pipeline validation only after P1 and pelvis are expressed in the same frame.
Real A3 bridge	Pelvis IMU plus joint state; no checked-in absolute translation source.	A temporary independent 6-DOF reference must be integrated for real calibration.
HOPE runtime TF	world -> P1 -> pelvis_link.	Provides absolute root localization to ROS consumers after calibration.

A TF named P1 -> pelvis_pose would not solve the missing-reference problem. pelvis_pose is a message stream; pelvis_link is the physical frame.

Frame alignment reminder

The MuJoCo pose topic is expressed in odom, while arena mocap normally uses world. They may be compared only after a known world/odom alignment or after both inputs are represented in one common frame.

Go/no-go checklist

A successful calibration is not just a low residual number. Use these high-level checks before accepting the JSON for competition.

Independent reference
The full pelvis pose comes from a source that does not depend on P1 or the static transform being measured.

Common reference frame
Both poses are world-referenced, or both have been transformed into another explicitly shared frame.

Synchronized acquisition time
Timestamps represent compatible acquisition times, not merely arrival time on two unrelated paths.

Sufficient motion
The pelvis moves through meaningful translation and rotation; a stationary capture is rejected.

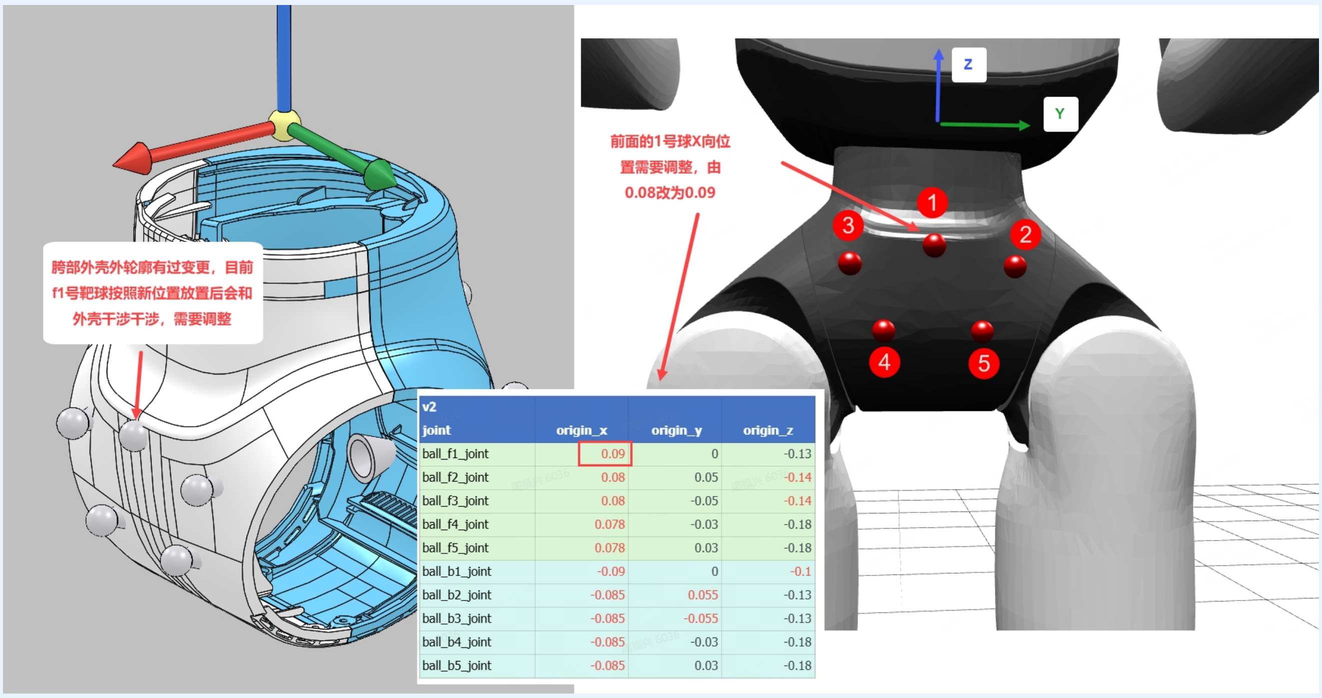
Traceable saved result
The JSON is associated with the exact robot, marker shell, Motive profile, and calibration session.

Single correction path
Either Motive applies the pivot correction or ROS publishes the static transform - never both.

Runtime acceptance: world -> P1 moves with the robot; P1 -> pelvis_link remains constant.

Identify the correct PKU v2 ten-marker shell

Match the shell and f1-f5 / b1-b5 positions before defining P1 in Motive.



v2 joint	origin_x	origin_y	origin_z
ball_f1_joint	0.09	0	-0.13
ball_f2_joint	0.08	0.05	-0.14
ball_f3_joint	0.08	-0.05	-0.14
ball_f4_joint	0.078	-0.03	-0.18
ball_f5_joint	0.078	0.03	-0.18
ball_b1_joint	-0.09	0	-0.1
ball_b2_joint	-0.085	0.055	-0.13
ball_b3_joint	-0.085	-0.055	-0.13
ball_b4_joint	-0.085	-0.03	-0.18
ball_b5_joint	-0.085	0.03	-0.18

PKU v2 reference:agibot/pku/hip_marker_shell/hip_marker_layout_and_coords_v2.png

The revised front f1 position is X = 0.09 m. Use all ten points shown; the live 6-DOF calibration remains authoritative.

Technical references

This guide intentionally stays at the system and operator level. Use the following repository documents for commands, parameters, message types, quality thresholds, and troubleshooting.

Repository overview
README.md
See 'Required mocap bringup: align P1 to A3 pelvis_link' for the shortest supported workflow.

Authoritative mocap contract
mocap/README.md
See 'Calibrating a humanoid P1 body to pelvis_link' for topic and frame semantics shared by the rest of HOPE.

OptiTrack operating procedure
docs/OPTITRACK.md
See 'Calibrating P1 to an A3 pelvis_link' for NatNet setup, calibration commands, validation checks, JSON output, and runtime publication.

Coordinate-frame contract
docs/interfaces/frames.md
Defines the shared HOPE world frame and the robot root-frame conventions.

Decision summary

Calibrate and save P1 -> pelvis_link.
Treat /a3/calibration/pelvis_pose as a temporary independent pose topic, not a TF frame.
At runtime, publish world -> P1 -> pelvis_link.

Generated from the current nightly_built documentation. If implementation and narrative ever diverge, follow the checked-in code and docs/ interface contracts.